

RAAHE LAPALUOTO, Finland

WMO: 028720

Lat: 64.6658N

Lon: 24.4069E

Elev: 2

StdP: 101.30

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.2	-20.0	-26.0	0.4	-23.1	-22.6	0.5	-19.9	18.7	-0.3	17.6	-0.5	3.2	70	0.808

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.1	24.4	18.5	22.3	17.7	20.6	16.7	19.8	23.1	18.5	21.4	17.3	20.0	4.5	10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.6	13.4	21.5	17.2	12.3	20.2	16.0	11.4	19.1	56.6	23.1	52.3	21.7	48.6	20.0	24.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.6	13.8	12.2	DB	-25.0	27.8	3.3	3.2	-27.4	30.1	-29.3	32.0	-31.1	33.8	-33.4	36.2
			WB	-24.5	20.9	4.0	2.2	-27.4	22.5	-29.7	23.8	-32.0	25.0	-34.9	26.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	4.1	-7.6	-6.5	-3.7	1.2	8.0	12.6	16.5	15.0	11.1	4.6	0.5	-3.0
	DBStd	9.23	6.29	6.70	4.80	3.07	4.32	3.40	3.28	2.86	2.86	3.25	4.11	5.68
	HDD10.0	2679	547	462	424	264	93	12	1	1	21	169	283	404
	HDD18.3	5222	805	695	682	513	323	177	77	112	217	426	533	662
	CDD10.0	531	0	0	0	1	29	88	202	154	55	1	0	0
	CDD18.3	32	0	0	0	0	2	3	21	7	0	0	0	0
	CDH23.3	127	0	0	0	0	7	15	80	25	0	0	0	0
	CDH26.7	29	0	0	0	0	1	4	19	5	0	0	0	0
	WSAvg	6.0	6.3	6.4	6.4	5.5	5.7	5.7	4.9	5.5	5.9	6.4	6.6	7.1
	PrecAvg	544	38	30	27	28	42	51	71	62	52	51	48	43
Precipitation	PrecMax	666	68	57	57	66	81	111	131	114	102	87	94	94
	PrecMin	405	10	7	5	10	13	5	15	17	19	5	8	13
	PrecStd	78	17	14	13	14	19	28	29	28	25	23	20	19
	PrecStd	78	17	14	13	14	19	28	29	28	25	23	20	19
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.9	3.6	6.2	12.4	24.2	25.1	29.5	26.5	20.2	12.6	8.5	5.3
		MCWB	2.3	2.5	2.6	9.2	14.4	18.2	19.3	19.7	15.3	10.9	7.2	4.4
	2%	DB	1.9	2.7	3.7	9.0	19.9	21.5	25.2	22.5	17.6	11.1	7.5	4.1
		MCWB	1.1	1.6	2.0	5.9	13.1	15.8	19.4	18.4	14.3	9.6	6.6	3.0
	5%	DB	1.3	1.9	2.9	6.6	16.6	19.4	23.3	20.5	16.0	9.8	6.3	3.2
		MCWB	0.7	1.0	1.4	4.2	11.9	15.0	18.6	17.2	13.6	8.5	5.5	2.3
	10%	DB	0.4	1.1	2.0	5.3	14.2	17.8	21.6	19.1	14.9	8.9	5.3	2.4
		MCWB	-0.1	0.4	0.8	3.3	10.1	14.1	17.8	16.3	13.0	7.7	4.5	1.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.5	2.8	3.6	10.0	15.5	18.8	21.9	20.7	16.1	11.5	7.6	4.5
		MCDB	2.7	3.5	5.4	12.0	21.1	22.6	24.6	24.5	18.9	12.2	8.1	5.0
	2%	WB	1.3	1.8	2.3	6.0	13.9	16.9	20.5	19.1	15.0	10.0	6.7	3.2
		MCDB	1.7	2.5	3.4	8.6	18.8	20.2	23.7	22.1	16.8	10.8	7.4	3.8
	5%	WB	0.7	1.2	1.6	4.6	12.2	15.6	19.3	17.7	14.1	8.7	5.6	2.4
		MCDB	1.1	1.8	2.6	6.2	16.5	18.8	22.5	20.0	15.4	9.5	6.2	3.0
	10%	WB	0.0	0.5	1.0	3.5	10.6	14.4	18.1	16.6	13.4	7.9	4.6	1.7
		MCDB	0.5	1.1	1.8	5.0	13.6	17.3	21.1	18.7	14.7	8.8	5.2	2.3
Mean Daily Temperature Range	5% DB	MDBR	5.2	5.0	5.4	4.8	5.6	5.2	5.1	5.1	4.5	3.9	3.3	3.8
		MCDBR	3.1	3.5	4.1	7.0	9.7	8.5	8.4	7.4	6.8	4.3	3.4	3.1
		MCWBR	3.1	3.1	3.1	4.8	5.4	4.8	4.6	4.5	4.5	3.7	3.4	2.8
	5% WB	MCDWR	3.4	3.5	3.8	6.6	8.9	7.2	7.3	6.2	5.4	4.1	3.3	3.0
		MCWBR	3.4	3.2	3.1	4.9	5.4	4.6	4.6	4.3	4.0	3.7	3.4	2.8
		MCWBR	3.4	3.2	3.1	4.9	5.4	4.6	4.6	4.3	4.0	3.7	3.4	2.8
Clear-Sky Solar Irradiance	taub		0.213	0.279	0.308	0.346	0.347	0.351	0.370	0.355	0.332	0.310	0.343	0.352
	taud		1.967	2.115	2.163	2.222	2.424	2.479	2.446	2.480	2.525	2.362	2.539	2.585
	Ebn at Noon		313	605	762	820	862	866	834	814	758	586	286	130
	Edh at Noon		25	61	90	108	98	95	96	84	66	49	22	13
All-Sky Solar Radiation	RadAvg		0.12	0.65	2.12	3.78	4.98	5.57	5.12	3.82	2.13	0.77	0.17	0.04
	RadStd		0.01	0.13	0.22	0.31	0.48	0.46	0.47	0.36	0.22	0.13	0.03	0.00

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (39 neighbors)	+0.48	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page